Likes or Dislikes, Gratifications or Concerns? A One-Week Field Experiment Analyzing the Effects of Popularity Cues on the Privacy Calculus in Online Communication

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Abstract

The privacy calculus model states that people weigh risks and benefits before communicating online. This paper analyzes three questions: (1) How robust is the privacy calculus, can it be replicated with actual behavior in realistic online settings? (2) Do we need to extend the model by integrating more socio-psychological measures such as trust, self-efficacy, and privacy deliberation? (3) How strongly is the privacy calculus affected by affordances such as like and dislike buttons? In a preregistered one-week field experiment (N = 590), participants discussed a current political topic on an online website, which included either (a) like buttons, (b) both like and dislike buttons, or (c) no buttons. **The privacy calculus model was confirmed: Benefits and concerns affected communication. Deliberating about privacy by comparing benefits and risks decreased communication, whereas experiencing self-efficacy and trust increased communication. Although like and dislike buttons did not affect concerns, experiences, trust, efficacy, and deliberation, they had a strong effect on behavior: They reduced communication by 45 percent.**

Keywords: privacy calculus, communication, popularity cues, field experiment, preregistration, preregistration

Word count: 8,432

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Introduction

According to the privacy paradox, the way users share information online is erratic (Barnes, 2006): Although strongly concerned about their privacy, people communicate much personal information online (Taddicken & Jers, 2011). However, despite its popularity in both academic research (Masur, 2023) and the public press (New York Public Radio, 2018), empirical support for the privacy paradox is tenuous (Baruh et al., 2017). **As an alternative explanation**, research is increasingly building on the privacy calculus model (Dienlin, 2023), which states that communication online can be explained by perceived risks and expected benefits (Bol et al., 2018; Krasnova et al., 2010).

Although the privacy calculus has gained momentum in academic research, several important questions remain unanswered. First, the privacy calculus' empirical foundation needs to be improved (Knijnenburg et al., 2017). To date, much research on the privacy calculus builds on (a) self-reports of behavior (Krasnova et al., 2010), (b) vignette approaches (Bol et al., 2018), or (c) one-shot experiments in the lab (Trepte et al., 2020). Self-reports are often unreliable measures of online behavior (Parry et al., 2021), vignette approaches have limited external validity, and one-shot experiments cannot analyze effects of sustained use. The first aim is therefore to replicate the privacy calculus model in a more authentic and long-term setting using actual behavioral data. Doing so will show its robustness, and it will help understand if behavior is better explained by the privacy paradox approach or by the privacy calculus model.

Second, the privacy calculus model is criticized for neglecting the actual process of weighing **benefits and costs** and for over-emphasizing rationality (Knijnenburg et al., 2017). According to critics, simply showing that both concerns and gratifications correlate with communication online does not prove that an explicit weighing process took place. Instead, we would need to find out if people deliberately compare costs and benefits before communicating online. Therefore, we will also analyze the process of comparing **benefits and costs** explicitly,

drawing on the privacy deliberation process described by Omarzu (2000). Next, focusing exclusively on costs and benefits might not fully capture how and why users engage online. We, hence, extend the privacy calculus by analyzing how online communication is affected by trust and self-efficacy, two variables less focused on rationality and more oriented toward socio-psychological aspects critical in online contexts (Metzger, 2004).

Online behavior is determined by the person and their external circumstances (Bazarova & Masur, 2020; Spottswood & Hancock, 2017; Trepte et al., 2020). **Implies moderation, in line with reviewer comment. Moderation analyses added to companion "Moderation Analyses": The privacy calculus upholds that much of behavior is rational and self-determined, but how strongly does this process depend on online affordances? Update Fig 1?** It is well known that the affordances of many online services are optimized to elicit as much interaction as possible (Ellison & Vitak, 2015; Masur et al., 2021), for example via low threshold communication features such as likes, shares, replies, or reactions (Carr et al., 2018). Implicit and explicit cues on how a website is used can increase communication (Spottswood & Hancock, 2017; Trepte et al., 2020). How large are the effects of external popularity cues such as like and dislike buttons compared to the internal weighing of benefits and costs? And how easily can the mechanisms of the privacy calculus be affected by means of popularity cues?

Update In conclusion, in this paper we contribute to communication theory by means of (a) replication, (b) falsification, (c) extending the range of existing theory, (d) elucidating mechanisms, and (e) theory comparison (DeAndrea & Holbert, 2017).

Replicating the Privacy Calculus

The privacy calculus analyzes why people communicate online. When are we willing to engage in a conversation? It builds on the calculus of behavior (Laufer & Wolfe, 1977) and states that people are weighing risks and benefits before actively communicating. Communication is closely related to privacy, which is defined as a "voluntary and temporary withdrawal of a person from the general society" (Westin, 1967, p. 7). People regulate their privacy by deciding what and what not to communicate (Dienlin, 2014). Communication is therefore also closely related to

self-disclosure. Just as it is impossible to not communicate, communication inherently promotes self-disclosure. In a recent study, communication quantity and the frequency of expressing one's political opinion was almost indistinguishable (r = .91; **AUTHORS**). Next to breadth and depth, communication quantity is hence often considered a central dimension of self-disclosure (Omarzu, 2000).

Sharing information carries risks, as recipients may reuse it in different contexts, potentially harming the original sender (Masur et al., 2021). As a result, people who aim to avoid risks should be inclined to share less information, especially in online contexts where audiences are much larger (Vitak, 2012). Indeed, empirical research confirms that people with higher privacy needs and privacy concerns are less likely to communicate online (Krasnova et al., 2010; Masur, 2023; Masur & Trepte, 2021). This finding is now confirmed by several meta-analyses and reviews (Baruh et al., 2017; Dienlin & Sun, 2021; Gerber et al., 2018). Similarly, people who are more concerned about their privacy also engage in more privacy protection behaviors (Baruh et al., 2017; Stubenvoll et al., 2022).

Even outweighing concerns, the most relevant drivers of online communication are expected gratifications (Bol et al., 2018; Dienlin & Metzger, 2016; Kezer et al., 2022). On average, people are happy to trade in parts of their privacy to receive something more valuable in return (Laufer & Wolfe, 1977). In online communication, the most important benefits include social support, social capital, entertainment, information-seeking, and self-presentation (Dhir & Tsai, 2017; Ellison et al., 2007; Krasnova et al., 2010; Whiting & Williams, 2013).

H1: People who are more concerned about their privacy are less likely to communicate actively on a website.

H2: People who obtain more gratifications from using a website are more likely to communicate actively on a website.

Extending the Privacy Calculus

Although privacy calculus implies that people explicitly compare benefits and risks before communicating online, prior research has neglected this aspect (Knijnenburg et al., 2017). Only

observing that privacy concerns or expected gratifications and communication online are related does not prove that an explicit weighing process took place (Knijnenburg et al., 2017). Instead, we need to analyze if, and if so by how much, people actively deliberate about their privacy by comparing benefits and risks, and whether doing so influences their willingness to communicate. Self-disclosure theory (Altman, 1976; Omarzu, 2000) suggests that if the benefits of communication are attractive, deciding whether or not to communicate is a "conscious and deliberative process" (Omarzu, 2000, p. 183). **We hence extend the privacy calculus by introducing a measure of privacy deliberation.** Privacy deliberation captures the extent to which individuals explicitly compare risks and benefits before communicating with others (**AUTHORS**).

How could deliberating about one's privacy affect communication? On the one hand, it could reduce subsequent communication. Refraining from communication—the primary means of connecting with others (Altman, 1976)—often requires restraint (Omarzu, 2000). This is especially true for social media, which are designed to foster communication and participation (Ellison & Vitak, 2015; Masur et al., 2021). Actively thinking about whether communicating is worthwhile might be the first step not to participate. In addition, actively reflecting about one's behavior represents a central and Type 2 approach toward decision making, which is often associated with more critical and cautious behavior (Kahneman, 2011; Petty & Cacioppo, 1986).

On the other hand, deliberating about privacy might also increase communication. The default behavior in online contexts is passively browsing the content, but not actively engaging in communication (Ozimek et al., 2023). Especially in new contexts without prior experience, actively pondering one's options might trigger users to leave their default state of passiveness and to become active and involved. In light of the numerous benefits mentioned above, it might make sense to conclude that participation is beneficial, thereby fostering communication (Krasnova et al., 2010).

RQ1: Do people who deliberate more actively about their privacy communicate more or less **often** online?

It is useful to understand the privacy calculus from the perspective of *bounded rationality* (Simon, 1990). Bounded rationality states that "(1) humans are cognitively constrained; (2) these constraints impact decision making; and (3) difficult problems reveal the constraints and highlight their significance" (Bendor, 2015, p. 1303). It is important to emphasize that although human behavior is considered *partly* irrational, bounded rationality does not state that it is *completely* irrational (Gigerenzer et al., 2002). Instead, rationality needs to be understood as a continuum. And in the context of online privacy, rationality is impeded by information asymmetries, presence bias, intangibility, illusory control, or herding (Acquisti et al., 2020). It follows that to provide a more complete picture, additional factors less focused on rationality but more on socio-psychological aspects should also explain communication.

Two central factors that help us understand online communication are self-efficacy and trust (Hossain & Wigand, 2004; Metzger, 2004). Privacy violations create psychological distress (Ledbetter, 2019). Experiencing online contexts as a safe space that users can sufficiently control is important for engaging in online communication. If users are more familiar, experienced, and knowledgeable in a given online context, they are more likely to navigate that online context successfully and to communicate actively (Baruh et al., 2017; Krämer & Schäwel, 2020; Park, 2013). People with more privacy self-efficacy engage more successfully in self-withdrawal (Dienlin & Metzger, 2016) and protective online behavior (Boerman et al., 2021; van Ooijen et al., 2022). Hence, if users possess more self-efficacy to participate, they should also communicate more.

H3: People are more likely to communicate on a website when their self-efficacy to actively use the website is higher.

In all situations where people lack experience, control, or competence, a central variable to understand behavior is trust (Gefen et al., 2003). Trust plays a key role especially in online contexts (Metzger, 2004, 2006). Users often cannot control the environment or the way their information is handled (Acquisti et al., 2020; Bräunlich et al., 2020). Trust either captures "*specific* beliefs dealing primarily with the integrity, benevolence, and ability of another party"

(Gefen et al., 2003, p. 55, emphasis added) or a "*general* belief that another party can be trusted" (Gefen et al., 2003, p. 55, emphasis added). In online contexts, there are different targets of trust, including (a) the information system, (b) the provider, (c) the Internet, and (d) the community of other users (Söllner et al., 2016). People who put more trust in the providers of networks, for example, disclose more personal information (Li, 2011). To comprehensively capture and understand online communication, trust should hence be included.

H4: People are more likely to communicate on a website when they have greater trust in the provider, the website, and the other users.

Analyzing the Impact of Popularity Cues

How are the privacy calculus and communication affected by the context, the digital infrastructure? How easily can the calculus be manipulated externally? One of the central tools to afford and govern online behavior are popularity cues such as like and dislike buttons (Stsiampkouskaya et al., 2023). Popularity cues have been shown to affect behavior (Krämer & Schäwel, 2020; Masur et al., 2021; Trepte et al., 2020). For example, online comments that already have several dislikes are more likely to receive further dislikes (Muchnik et al., 2013). When users disagree with a post, they are more likely to click on a button labeled *respect* compared to a button labeled *like* (Stroud et al., 2017).

How and why might popularity cues affect the privacy calculus? In analyzing this question, it makes sense to analyze the cues' underlying affordances (Ellison & Vitak, 2015; Fox & McEwan, 2017). Affordances are mental representations of how objects might be used. They emphasize that it is not the *objective features* that determine behavior, but rather our *subjective perceptions* (Gibson, 2015). Popularity cues such as like and dislike buttons, which are of primary interest in this study, are understood as "paralinguistic digital affordances" (Carr et al., 2018, p. 142), lowering thresholds to partake in online communication.

Popularity cues likely impact the privacy calculus via two underlying theoretical mechanisms (Krämer & Schäwel, 2020): First, the *mere presence* of popularity cues might affect whether people are willing to disclose. Being able to attract likes might afford active

communication. The mere option to receive dislikes, conversely, might inhibit communication. Second, *_actually receiving* likes or dislikes might affect behavior, by means of positive reinforcement (likes) or punishment (dislikes) (Skinner, 2014). To illustrate, likes are affirmative and embody the positivity bias of social media (Schreurs et al., 2022). Receiving a like online is similar to receiving a compliment offline (Carr et al., 2018; Sumner et al., 2017). Like buttons afford and emphasize a *gain frame* (Rosoff et al., 2013). These gains can be garnered only through active participation. In situations where people can gain immediate positive outcomes, concerns and risks that are more vague and in the future often become less relevant (presence bias, Ainslie & Haslam, 1992). Because like buttons emphasize positive outcomes, it is likely that concerns decrease. In situations where there is more to win, people might also more actively deliberate about whether or not to communicate.

Dislikes, instead, represent a punishment, introducing a *loss frame*. Websites featuring both like *and* dislike buttons should therefore be more ambivalent compared to websites without popularity cues, fostering privacy deliberation. Privacy concerns should not be reduced anymore: People who need more privacy are also more shy and risk averse (Dienlin & Metzger, 2024). Implementing the dislike button might therefore increase privacy concerns, canceling out the positive effects of the like button. At the same time, communication and benefits might still be increased compared to a website without like and dislikes buttons, as online benefits are often considered to outweigh risks (positivity bias, Schreurs et al., 2022).

H5: Compared to a control group with no like and dislike buttons, people who use a website with like buttons (a) communicate more; (b) obtain more gratifications; (c) are less concerned about their privacy; and (d) deliberate more about whether they should communicate online.

H6: Compared to a control group with no like and dislike buttons, people who use a website with like *and* dislike buttons (a) communicate more; (b) obtain more gratifications; and (c) deliberate more about whether they should communicate online.

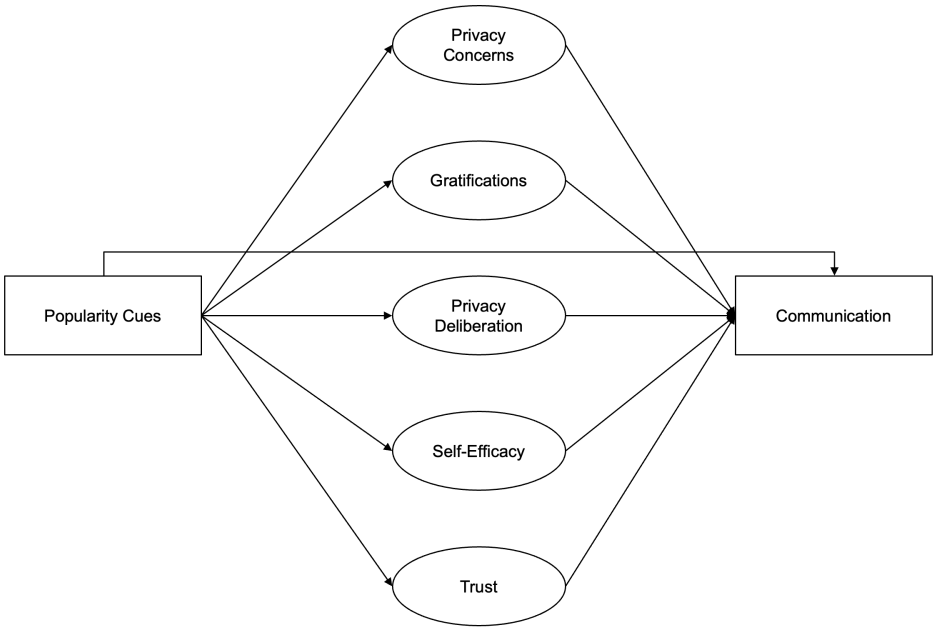
H7: Compared to people who use a website with only like buttons, people who use a

website with like and dislike buttons (a) are more concerned about their privacy, and (b) deliberate more about whether they should communicate online.

In conclusion, this leads to an updated and extended model of the privacy calculus, in which the affordances of online contexts affect communication mediated via experienced privacy concerns, expected gratifications, privacy deliberation, self-efficacy, and trust. For an overview of the theoretical model, see Figure 1 **Update Figure?**.

Figure 1

Overview of the extended privacy calculus model.



Methods

Open Science

This manuscript features a companion website, which includes the data, research materials, analysis scripts, and a reproducible version of this manuscript. The hypotheses, sample size, research materials, analyses, and exclusion criteria were preregistered (see on OSF). In some cases, the preregistered approach had to be changed (see *Open Science and Preregistration* on the companion website).

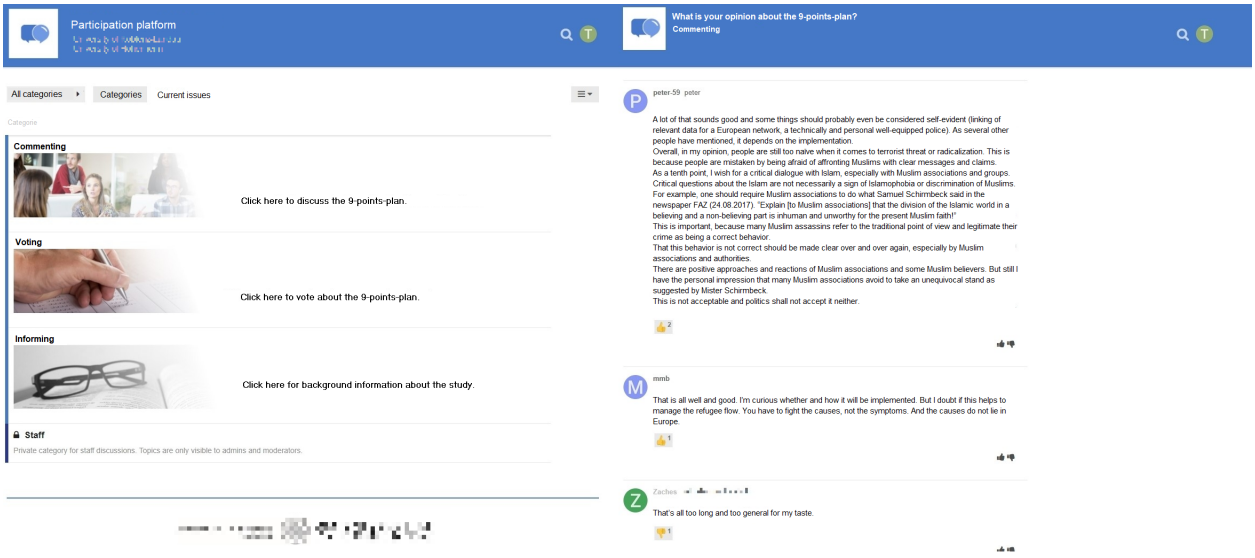
Procedure

The study was designed as an online field experiment with three different groups. The first experimental group used a website that included like buttons; the second experimental group used an identical website including both like and dislike buttons; and the control group used an identical website without like and dislike buttons. Participants were randomly distributed to one of the three websites in a between-subject design. The data were collected in Germany. Participants were recruited using the professional panel agency Norstat. **Norstat allowed us to draw a quota sample matching the German population in age, gender, and federal state. We collected the data in Germany because the website, topic, and materials were designed in German.** As incentive, participants were awarded digital points, to receive special offers from online retailers. Participants had to be over 18 years old and reside in Germany. In a first step, the company sent its panel members an invitation to participate in the study. In this invitation, panel members were asked to participate in a study analyz-ing the current threat posed by terrorist attacks. Members who agreed to participate were sent the first questionnaire (*T1*). The questionnaire asked participants about their sociodemographic background, (b) provided more details about the study, and (c) included a registration link for the website, which was introduced as "participation platform". Afterward, participants were randomly assigned to one of the three websites. After registration was com-pleted, participants were invited (but not obliged) to visit the website and to discuss the topic of the terrorism threat in Germany. **We chose terrorism as the discussion topic because it attracts strong media attention and shapes mass media debates (Völker, 2024). Germany has also been among the Western European countries most affected by right-wing terrorism and violence (Ravndal et al., 2021), and Islamist attacks are highly prevalent and visible in the media. The topic is also well suited to our content analysis: it links to personal attitudes, which is useful for coding valence, and its salience and personal relevance elicit personal experiences.** Participants could use the website and write comments over the course of one week. **We chose one week to allow sustained use of the platform beyond a one-shot exposure, while limiting dropout and the risk that external**

events would change the discussion during the field phase. Subsequently, participants received a follow-up questionnaire in which the self-reported measures reported below were collected (*T2*).

Figure 2

Screenshot of the landing page and of communication taking place. Translated into English.



The online website was programmed based on the open-source software discourse. To make sure the website was professional and authentic, several pretests with students from the local university were run. Nine hundred sixty participants created a user account on the website (see below) and used the website actively. Overall, they spent 162 hours online, wrote 1,171 comments, and clicked on 560 popularity cues. All communication was checked, and there were no instances of people providing meaningless text or doubting the experiment. For a screenshot of the landing page and for examples of comments, see Figure 2.

Power Analysis

Sample size was determined using a priori power analyses. The power analyses were based on estimates from the literature. When researching aspects of privacy online (e.g., Baruh et al., 2017), effects are often small (i.e., r = .10, Cohen, 1992). Hence, the minimum effect size was set to be r = .10. The aim was to be able to detect this effect with a probability of at least 95% (i.e., power = 95%). Using the regular alpha level of 5%, power analyses suggested a minimum sample size of N = 1,077. In the end, we were able to include N = 559 in the analyses (see below),

which was significantly lower than the original aim. With this sample size, the study had a power of 77% to find an effect at least as large as $r = .10$. Sensitivity analyses showed that the current study could still make reliable results (i.e., with a power of 95%) for effects at least as large as $r = .14$. In conclusion, although not as powerful as planned, the study is still adequately powered to find the small effects reported in the privacy literature (Baruh et al., 2017).

Participants

A quota sample that matched the German population in terms of age, gender, and federal state was collected. In sum, 1,619 participants completed the survey at T1; 960 participants created a user account on the website; and 982 participants completed the survey at T2. The data were connected using tokens and IP addresses. For technical reasons, the data of several participants could not be matched (for example, because they used different devices for the respective steps). In the end, the data of 590 participants could be matched successfully. Considered unreasonably fast, 29 participants were excluded who finished the questionnaire at T2 in less than three minutes. To detect atypical data and response patterns, Cook's distance was calculated. We excluded 2 participants with clear response patterns (i.e., straight-lining). The final sample included $N = 559$ participants. The sample characteristics at T1 and T2 were as follows. T1: age = 45 years, gender = 49% male, college degree = 22%. T2: age = 46 years, gender = 49% male, college degree = 29%. One participant did not report their gender.

Measures

Factor validity was assessed using confirmatory factor analyses (CFA). If the CFAs revealed insufficient fit malfunctioning items were deleted. All items were measured on bipolar 7-point scales. Answer options were visualized as follows: -3 (*strongly disagree*), -2 (*disagree*), -1 (*slightly disagree*), 0 (*neutral*), +1 (*slightly agree*), +2 (*agree*), +3 (*strongly agree*). For the analyses, answers were coded from 1 to 7. All items measuring the same variable were presented in randomized order on the same page.

All measures showed high factorial validity. For an overview of the means, standard deviations, factorial validity, and reliability, see Table 1. For the variables' distributions, see

Table 1
Psychometric Properties, Factorial Validity, and Reliability of Measures

	m	sd	chisq	df	p	cfi	tli	rmsea	srmr
Privacy concerns	3.21	1.51	25.84	9	0.00	1.00	0.99	0.06	0.01
Gratifications general	4.76	1.22	53.09	5	0.00	0.98	0.96	0.13	0.02
Privacy deliberation	3.93	1.29	27.42	5	0.00	0.98	0.96	0.09	0.02
Self-efficacy	5.25	1.12	10.53	1	0.00	0.99	0.95	0.13	0.01
Trust general	5.21	1.04	1.86	1	0.17	1.00	1.00	0.04	0.01
Trust specific	5.08	0.94	153.12	24	0.00	0.96	0.94	0.10	0.04

Note. omega = Raykov's composite reliability coefficient omega; avevar = average variance extracted.

Figure 3. For all items and their distributions, see companion website.

Privacy concerns were measured with seven items based on Buchanan et al. (2007). One example item was "When using the participation platform, I had concerns about my privacy". One item was deleted due to poor psychometric properties. The mean was m = 3.21. This value is below the scale's midpoint of 4, showing that on average people were not strongly concerned about their privacy.

General gratifications were measured with five items based on Sun et al. (2015). One example item was "Using the participation platform has paid off for me". The mean was m = NA, which was above the scale's midpoint. This shows that on average people considered the website to be beneficial. For exploratory analyses, we also collected a scale measuring specific gratifications, not included here.

Privacy deliberation was measured with five self-designed items. One example item was "While using the participation platform I have weighed the advantages and disadvantages of writing a comment." The mean lay on the scale's neutral midpoint (m = 3.93). (For an interpretation, see below.)

Self-efficacy was captured with six self-designed items, which measured whether participants felt that they had sufficient self-efficacy to write a comment on the website. For example, "I felt technically competent enough to write a comment." Two inverted items were

deleted due to poor psychometric properties. People felt self-efficacious to use the website (m = 5.25).

Two types of trust were measured. *General trust* was operationalized based on Söllner et al. (2016), addressing three targets (i.e., provider, website, and other users), measured with one item each. One example item was "The operators of the participation platform seemed trustworthy." *Specific trust* was operationalized for the same three targets with three sub-dimensions each (i.e., ability, benevolence/integrity, and reliability), measured with one item each. Example items were "The operators of the participation platform have done a good job" (ability), "The other users had good intentions" (benevolence/integrity), "The website worked well" (reliability). Participants placed a lot of trust in the website, the providers and the other users (trust general: m = 5.21; trust specific: m = 5.08).

Communication was calculated by counting the number of words each participant wrote in a comment. Communication was zero-inflated and heavily skewed: While 58 percent did not communicate at all, the maximum number of words communicated by a single user was 3198 words. On average, participants wrote 77 words.

begin new

To operationalize self-disclosure, we follow Omarzu (2000) and distinguish breadth, depth, and communication quantity. Communication quantity was measured as the word count of each comment. Two research assistants manually coded a stratified random sample of 150 comments (50 per condition) for depth (personal experience, emotional expression, and political opinion), breadth (nine predefined topics), valence, and controversiality. We report high intercoder reliability (see Table 2).

The remaining comments were classified with qwen3.8:27b-q4_K_M, an open-source large language model run locally via Ollama with a few-shot system prompt and a temperature of 0. We choose it after comparing it with a pretrained zero-shot NLI classifier and other LLM few-shot classifiers (see Model Comparison) against the manually coded sample. Qwen3.8:27b-q4_K_M shows the highest agreement with the human coders (Cohen's κ = .66–.81,

Table 2
Intercoder Reliability

Variable	Categories	Agreement	K'dorff's α	Cohen's κ
Personal Experience	2	0.99	0.93	0.93
Emotional Expression	2	0.95	0.88	0.88
Political Opinion	2	0.95	0.90	0.90
Breadth	9	0.92	0.90	0.90
Valence	4	0.89	0.85	0.85
Controversy	2	0.93	0.83	0.83

Note. Categories = number of coding categories per variable. Agreement = proportion of identical codes. Cohen's kappa values of .40, .60, and .80 are commonly interpreted as moderate, substantial, and almost perfect agreement, respectively [@landisMeasurementObserverAgreement1977].

Krippendorff's α = .66–.90 across variables). Manually coded comments keep their human labels in all analyses. For the prompt and the full model comparison, see System Prompt and Model Comparison. Classification of user comments was performed using Python (Version 3.10+). All Python packages and their exact versions are listed in the file `requirements.txt` in the Github repository.

end new

Data Analysis

begin new

In this section, we present out modeling strategy. A more detailed versions of how we conducted this can be found on the companion website, see Analyses.

To analyze the association between the privacy calculus variables and platform affordances on one hand, and how they affect user communication on the other hand, we use Bayesian hurdle modeling. For ease of interpreation, logistic, OLS, and negative binomial regressions are provided on the companion website. We study two dependent variables: communication quantity and self-disclosure. For the detailed operationalization strategy, see Content Analysis Procedure on the companion website.

To model *communication quantity*, we rely on the number of words a user wrote in their

comments. Alternatively, we add the number of interactions they used (i.e., pressing the like or dislike button) to our measure. As a robustness check, we log transform both: the number of words variable as well as the interaction-extended variable. *Self-disclosure* is an index variable consisting of quantity, depth, and breadth (Omarzu, 2000).

To test our hypotheses, we use Bayesian hurdle models. The Bayesian hurdle model constitutes a two-step process, modeling a random variable. It is especially useful when a variable contains an excess of zeroes (i.e., it is zero-inflated and non-normal distributed). In a first step (the hurdle), the likelihood of the event happening at all (here: communication) is modeled using a logistic regression. Thereafter, a truncated count model estimates the effect size among the people who communicated. Whereas regular linear and logistic regression are easier to interpret, the Bayesian hurdle model is the modeling choice undertaken here because it takes the excess of zero values in the data into account. For more information on the fitted models, see online companion website.

The data were analyzed using R (Version 4.6.1; R Core Team, 2018) and the R-packages *brms* (Version 2.23.0; Bürkner, 2017, 2018), *lavaan* (Version 0.7.2; Rosseel, 2012), *marginaleffects* (Version 0.32.0; Arel-Bundock, 2023), *papaja* (Version 0.1.5; Aust & Barth, 2018), *pwr* (Version 1.3.0; Champely, 2018), *quanteda* (Version 4.5.0; Benoit, 2018), *semTools* (Version 0.5.9; Jorgensen et al., 2018), and *tidyverse* (Version 2.0.0; Wickham, 2017).

end new

Results

In this section, we present our findings. First, we describe the bivariate relations between all variables. Second, we use Bayesian hurdle models to analyze how the privacy calculus variables and the popularity cues (i.e., the availability of like and dislike buttons) are related to two outcomes: (1) communication, measured as the number of words participants wrote (preregistered), and (2) self-disclosure, an index of the quantity, breadth, and depth of what participants disclosed (exploratory).

Descriptive Analyses

We first plotted the bivariate relations of all variables (see Figure 3). All variables referring to the privacy calculus demonstrated the expected bivariate relationships with communication. For example, people who were more concerned about their privacy communicated less (r = -0.15). Self-disclosure was strongly related to communication (r = 0.87) and showed similar bivariate relations with the privacy calculus variables. The mean of privacy deliberation was m = 3.93. Altogether, 32% of participants reported having actively deliberated about their privacy.

The bivariate results showed two large correlations: specific trust and general gratifications (r = 0.79), and privacy concerns and privacy deliberation (r = 0.61).[1]

Bayesian Hurdle Model: Communication

Privacy Calculus

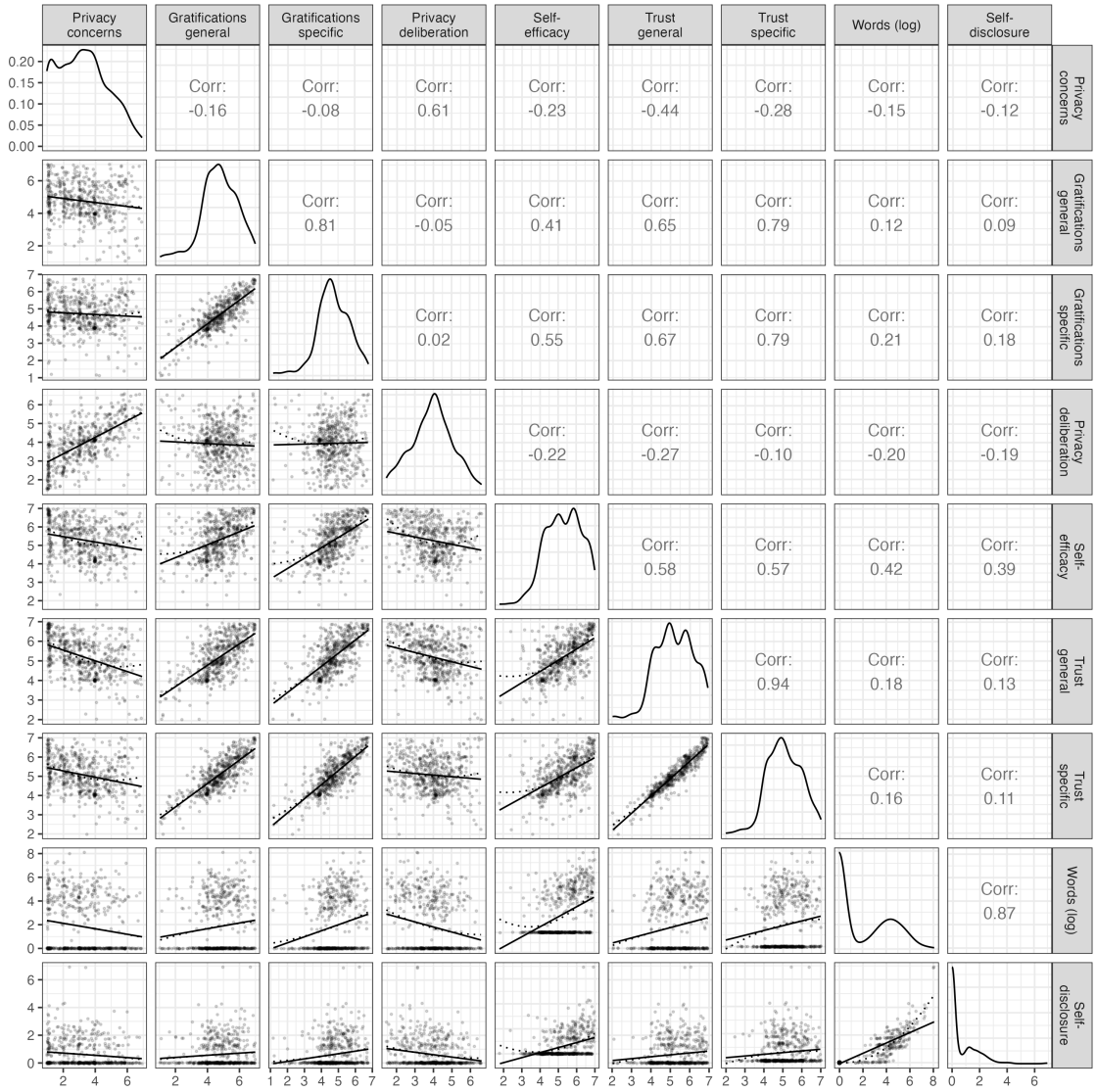
Hypotheses 1, 2, 3, and 4 were all supported. Participants who were more concerned about their privacy communicated less: With each one-SD increase in privacy concerns, participants wrote on average 24 fewer words (95% HDI [-41, -10]). Participants who expected more gratifications from participation communicated more: With each one-SD increase in expected gratifications, they wrote on average 31 more words (95% HDI [16, 49]). Participants who felt more self-efficacious communicated much more: With each one-SD increase in self-efficacy, they wrote on average 70 more words (95% HDI [54, 92]). The relationship was curvilinear, almost exponential: Starting from the lowest level of self-efficacy, an increase of one SD was associated with only 2 more words, whereas ending at the highest level, the same increase was associated with 147 more words.

Participants who placed more trust in the website, the provider, and the other users also communicated more: With each one-SD increase in trust, they wrote on average 30 more words (95% HDI [14, 50]). The first research question asked how privacy deliberation would be related to communication. The results revealed a negative relation: The more people deliberated about

[1] In the preregistered analysis, all six variables were analyzed within a single multiple regression. We report problems of occurring multicollinearity on the supplementary material on the companion website.

Figure 3

Above diagonal: zero-order correlation matrix; diagonal: density plots for each variable; below diagonal: bivariate scatter plots for zero-order correlations. Solid regression lines represent linear regressions, dotted regression lines represent quadratic regressions. Calculated with the model predicted values for each variable (baseline model).



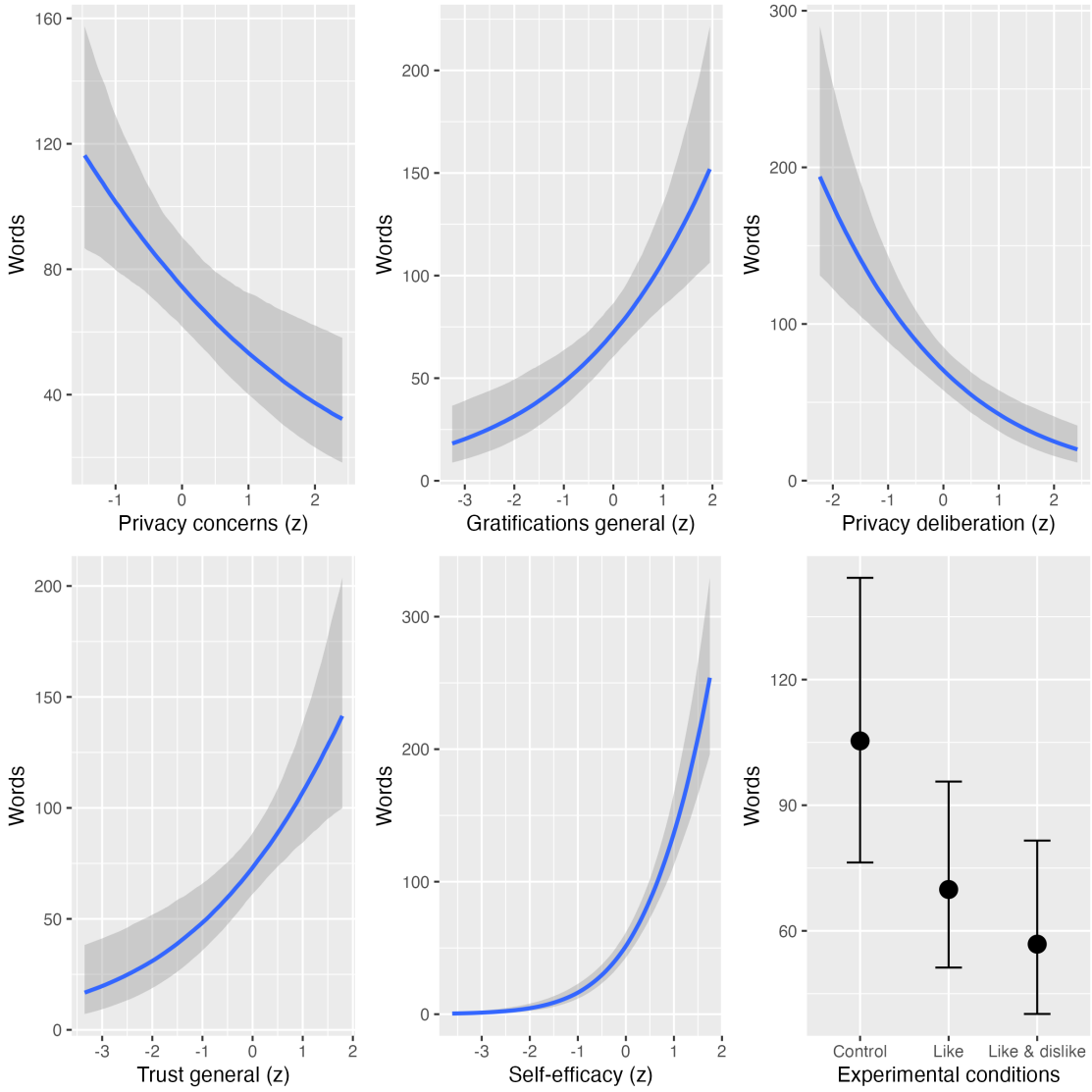
their privacy, the less they communicated. With each one-SD increase in privacy deliberation, participants wrote on average 37 fewer words (95% HDI [-58, -22]).

Popularity Cues

We then reanalyzed the effects of the popularity cues on communication. Compared to the control condition without popularity cues, implementing like buttons did not significantly affect

Figure 4

Bayesian hurdle models for communication (number of words). Plotted are the predicted values across the range of each predictor (privacy calculus variables standardized). The difference between the control and the like & dislike group is significant.



communication (b = -35, 95% HDI [-79, 2]), although the interval only narrowly included zero. However, implementing both like and dislike buttons affected communication. Contrary to what we expected, implementing both popularity cues *decreased* communication. If both popularity cues were present, participants wrote on average 48 fewer words (95% HDI [-91, -10]). The introduction of both cues hence led to a 45% decline in the number of words that were written.

Finally, we tested if the effect of the popularity cues on communication was potentially

mediated by the privacy calculus variables. Results showed that this was not the case. The popularity cues only affected behavior but not the predictors of the privacy calculus model (see online companion website). The results suggest that the effect was either direct or mediated by other variables not included here (Coenen, 2022).

Bayesian Hurdle Model: Self-Disclosure

In an exploratory step, we analyzed how the results change if we look not only at how much participants communicated, but at how much they disclosed about themselves. Self-disclosure combines the quantity (number of words), breadth (number of topics), and depth (personal experiences, emotional expression, and political opinions) of participants' comments (Omarzu, 2000). Participants who did not comment have a value of zero; all other values are expressed in standard deviations.

Privacy Calculus

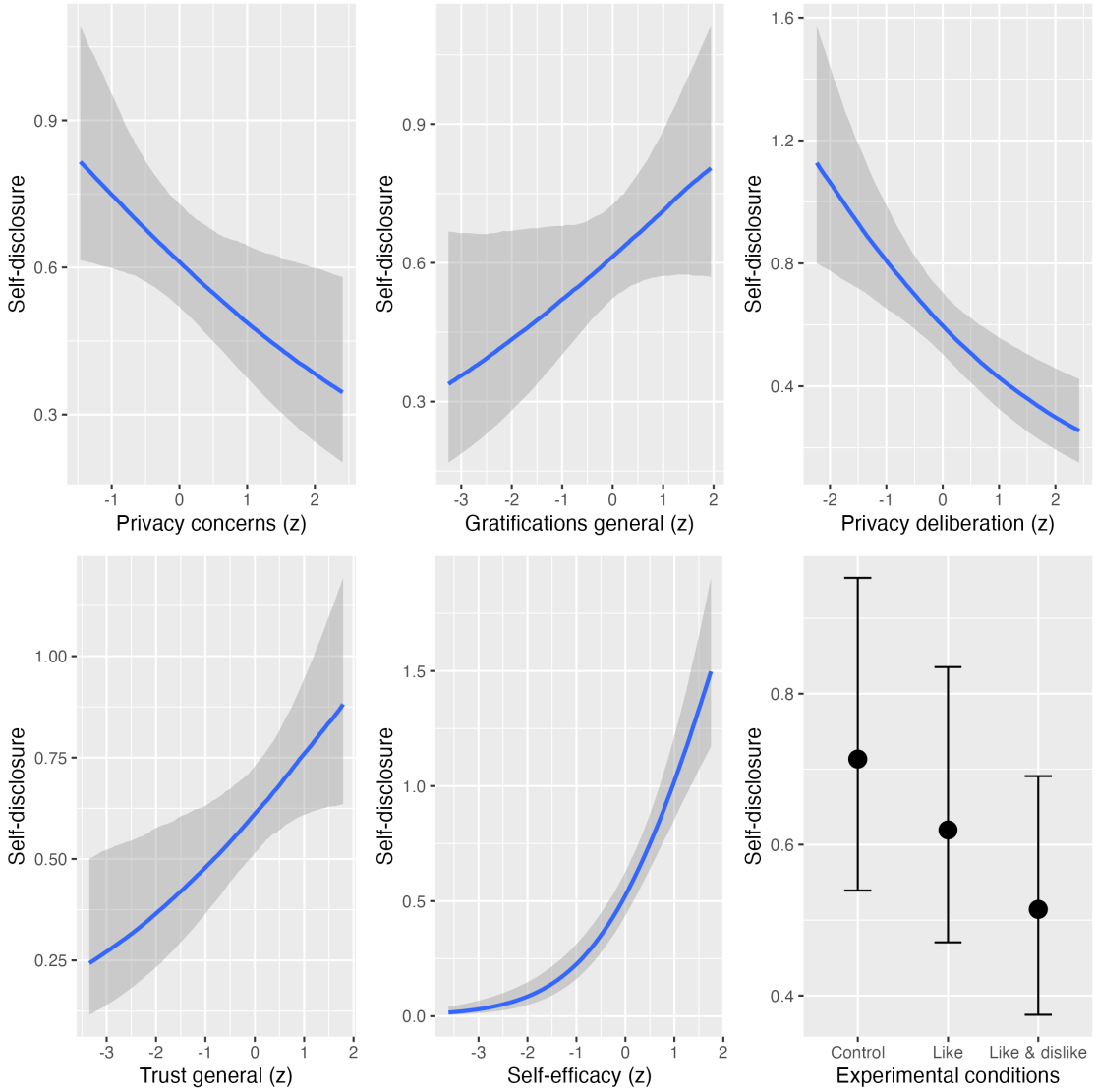
The results for self-disclosure largely mirrored those for communication. Participants who were more concerned about their privacy disclosed less (b = -0.13, 95% HDI [-0.24, -0.03]). In contrast to communication, expected gratifications were not significantly related to self-disclosure (b = 0.09, 95% HDI [-0.01, 0.2]). Participants who felt more self-efficacious disclosed more (b = 0.4, 95% HDI [0.3, 0.53]). Again, this relation was curvilinear: Starting from the lowest level of self-efficacy, an increase of one SD was associated with 0.03 SD more self-disclosure, whereas ending at the highest level, the same increase was associated with 0.63 SD more self-disclosure. Participants who placed more trust in the website, the provider, and the other users disclosed more (b = 0.14, 95% HDI [0.03, 0.26]), and participants who deliberated more about their privacy disclosed less (b = -0.19, 95% HDI [-0.3, -0.09]).

Popularity Cues

Compared to the control condition, implementing like buttons was not related to self-disclosure (b = -0.09, 95% HDI [-0.37, 0.18]). In contrast to communication, implementing both like and dislike buttons did not significantly reduce self-disclosure either (b = -0.2, 95% HDI [-0.47, 0.07]), although self-disclosure was descriptively 28% lower than in the control condition.

Figure 5

Bayesian hurdle models for self-disclosure. Plotted are the predicted values across the range of each predictor (privacy calculus variables standardized).



Taken together, the privacy calculus variables were related to self-disclosure in much the same way as to communication, with the exception of expected gratifications. The like and dislike buttons, however, primarily reduced how much participants wrote, and less so the breadth and depth of what they disclosed about themselves (see Table 3).

Table 3

Average marginal effects of the privacy calculus variables (standardized) and the popularity cues on communication and self-disclosure.

	Communication (words)			Self-disclosure (SD)		
Predictor	Estimate	LL	UL	Estimate	LL	UL
Experimental conditions						
Like vs. control	-35.00	-79.00	2.00	-0.09	-0.37	0.18
Like & dislike vs. control	-48.00	-91.00	-10.00	-0.20	-0.47	0.07
Like vs. like & dislike	13.00	-16.00	45.00	0.10	-0.15	0.34
Privacy calculus						
Privacy concerns	-24.00	-41.00	-10.00	-0.13	-0.24	-0.03
Expected gratifications	31.00	16.00	49.00	0.09	-0.01	0.20
Privacy deliberation	-37.00	-58.00	-22.00	-0.19	-0.30	-0.09
Trust	30.00	14.00	50.00	0.14	0.03	0.26
Self-efficacy	70.00	54.00	92.00	0.40	0.30	0.53

Note. HDI = highest density interval; LL = lower limit of the 95\% HDI; UL = upper limit. Intervals excluding zero are considered significant.

Discussion

This study analyzed three questions: Can the privacy calculus can be replicated with behavioral data in an authentic setting? Should the privacy calculus model be extended theoretically? Do like and dislike buttons affect online communication and the privacy calculus? To this end, a preregistered field experiment was conducted, which lasted one week. The privacy calculus model was extended: The privacy deliberation processes was tested explicitly, and trust and self-efficacy were included as predictors.

The preregistered analyses showed that the popularity cues did not affect communication. Only self-efficacy emerged as a significant predictor of online communication. All other variables remained insignificant. However, the preregistered analyses have to be treated with caution. The predictors were collinear, which makes their integration in one single model problematic (Vanhove, 2021). In addition, the main variable and outcome of the study, number of communicated words, was zero-inflated and gamma distributed, which requires a different type of analysis. The preregistered analyses using structural equation modeling with multiple predictors were hence problematic.

To address these issues, we conducted Bayesian hurdle-gamma models (see section Data analysis). This updated approach changed the results. People who were more concerned about their privacy wrote fewer words. To illustrate, people who reported being very much concerned posted only 33 words on average, whereas people who reported being not concerned posted 116 words. People who experienced privacy concerns hence differed strongly in their communication behavior from people who were unconcerned. Participants who received more gratifications wrote substantially more words. The effect was even larger, almost twice as large. For each point-increase in gratifications, participants wrote 27 words more. Attaining benefits online is hence a strong predictor of online communication. Together, the results provide further support for the privacy calculus and against the privacy paradox (Baruh et al., 2017). Communication online does not seem to be overly illogical. To large extents, it is aligned with respondents' concerns and benefits.

Results showed that trust and self-efficacy were important drivers of online communication. Participants who placed more trust into the website, the providers, and the other community members communicated more actively. Interestingly, self-efficacy was the strongest of all predictors. Participants who felt more self-efficacious disclosed much more than others. To illustrate, if people reported no self-efficacy, they wrote only 0 word on average. However, when they reported high levels of self-efficacy, they wrote 254 words. This finding further supports the underlying premise of bounded rationality (Simon, 1990). Although more rational aspects such as costs and benefits influence behavior, behavior is also determined by more socio-psychological variables such as trust and self-efficacy. The findings are therefore closely aligned with existing theory. For example, the technology acceptance model states that online behavior is most strongly determined by usefulness and ease of use (Venkatesh et al., 2003)—two variables closely related to gratifications and self-efficacy.

The privacy calculus was criticized for not explicitly analyzing the process of weighing pros and cons before disclosing (Knijnenburg et al., 2017). In this study, we thus elaborated on the privacy deliberation process. The results showed that only one third of all participants agreed to

have actively weighed the benefits and risks before communicating on the platform. This figure seems comparatively low. Even in new online contexts, the majority of users does not actively deliberate about their online communication, suggesting that online use is to large extents implicit (Acquisti et al., 2020).

Interestingly, and perhaps also somewhat surprisingly, privacy deliberation and privacy concerns were strongly correlated (r = .61). If we are concerned we also think and deliberate more actively about our privacy. And if we are not concerned we do not deliberate. This finding can be aligned with decision theory (Elsbach & Barr, 1999): When being concerned, we are in a negative state; and when in a negative state, we judge more critically. At this point, it is still unknown if thinking about privacy increases concerns or, conversely, if growing more concerned about privacy makes us deliberate more carefully.

The updated results showed that implementing both like and dislike buttons decreased communication. This was an unexpected and interesting finding. It suggests that negative feedback, or perhaps even only risks of negative feedback, can stifle communication. The effect was strong: Implementing both like and dislikes cues led to a 45% decrease in number of written words. When compared to the privacy calculus variables, we see that the effect is of similar size: Growing a bit more concerned or receiving less gratifications have a comparable impact on behavior. This finding is aligned with studies reporting strong effects of popularity cues on behavior (Muchnik et al., 2013).

The negative effects of dislike buttons might help explain why almost all existing successful social network sites have chosen to omit negative popularity cues. At the time of writing, only a handful of websites have (partially) implemented dislike buttons (e.g., youtube, stackexchange, or reddit). Despite the positivity bias of social media (Schreurs et al., 2022), chances of receiving negative feedback and communication are real, as can be seen by moral outrages or “shit-storms”. Explicit negative popularity cues are low threshold paralinguistic affordances (Carr et al., 2018). They likely prime or trigger negative experiences or expectations, thereby stifling communication. Interestingly, however, they did not affect the privacy calculus

variables, and no indirect effect was found. Hence, the effect is either transmitted via variables not included here (Coenen, 2022), or perhaps subconscious and direct.

Websites only including like buttons had no effect on the number of communicated words. If anything, there was an unexpected (non-significant) trend toward reduced communication. Although one might expect that like buttons, being positive feedback cues, increase communication, it is also plausible that they can decrease communication. Not receiving any likes can be perceived as ostracizing (Schneider et al., 2017). In the context of this study, participants discussed a political topic. Here, not receiving likes might be even more threatening and intimidating than on regular social media, where it is more common to discuss every-day and low-threshold topics. Although like buttons are commonplace in social media, the findings suggest that in specific contexts they inhibit communication.

Limitations

The main implications and results discussed above rest on exploratory analyses not registered a priori. Exploratory analyses are part of the research process, compatible with preregistration, and important for scientific progress. The updated analyses represent and document a learning process, which arguably led to an improved analysis. However, the results should still be considered somewhat preliminary, to be confirmed in subsequent studies.

Whereas the effects of the popularity cues on all variables can be interpreted from a causal perspective (but see below), more caution is needed regarding the effects of the privacy calculus variables on communication. Although the effects were controlled for age, gender, and education, other variables not included here could potentially bias the causal estimates (Coenen, 2022). In addition, in order not to reveal the study intention the self-reported measures were collected after the field phase. Demand effects might have led participants to align their answers to their prior behavior. To illustrate, users who communicated more actively might have experienced more self-efficacy afterward. As a result, the coefficients might overestimate the actual effects.

In experiments only the treatment variable should be manipulated, while all others should be held constant (assumption of stable unit treatment, Kline, 2016). Being a field experiment,

several variables could not be held constant, such as the content of communication by other users, the unfolding communication dynamics, and the characteristics of other users. Future research should repeat the design, preferably using several runs of the same experiment, to further assess generalizability and robustness.

Conclusion

This study provides further support for the privacy calculus model and against the privacy paradox approach. Expected benefits, privacy concerns, deliberating about benefits and risks, trust, and self-efficacy all affected communication. Like and dislike buttons reduced communication significantly. Users can be considered proactive and reasonable. But, similar to everyday offline contexts, they are also affected by the affordances of their environment, and often act implicitly without pondering the consequences of their actions.

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Appendix

Questionnaire Items

Response format

All items were measured on a 7-point scale with Likert-response format. Answers ranged between:

- -3 completely disagree
- -2 disagree
- -1 slightly disagree
- 0 neutral
- +1 slightly agree
- +2 agree
- +3 strongly agree

All items were translated from German.

Privacy concerns

Using the participation platform I had …

1. … concerns about what happens to my data.
2. … concerns about disclosing information about myself.
3. … no concerns.
4. … concerns that others could discover my real identity (i.e. my last and first name).
5. … concerns that information about myself could fall into wrong hands.
6. … concerns that others could discover what my political views are.
7. … concerns about my privacy.

Gratifications general

Using the participation platform …

1. … had many benefits for me.

2. … has paid off for me.
3. … was worthwhile.
4. … was fun.
5. … has brought me further regarding content.

Gratifications specific

Using the participation platform it has been possible for me …

Information

1. … to learn things I would not otherwise have noticed.
2. … to hear the opinion of others.
3. … to learn how other people tick.

Relevance

4. … to react to a subject that is very dear to me.
5. … to react to a subject that is important to me.
6. … to react to a subject that I am affected by.

Political participation

7. … to engage politically.
8. … to discuss political issues.
9. … to pursue my political interest.

Idealism

10. … to try to improve society.
11. … to advocate to something meaningful.
12. … to serve a good purpose.

Extrinsic benefits

13. … to do the responsible persons a favor.
14. … to soothe my guilty consciences.
15. … to fulfil my civic duty.

Privacy deliberation

Using the participation platform …

1. … I have considered whether I could be disadvantaged by writing a comment.
2. … I have considered whether I could be advantaged by writing a comment.
3. … I have weighed up the advantages and disadvantages of writing a comment.
4. … I have thought about consequences of a possible comment.
5. … I have considered whether I should write a comment or not.

Trust general

1. The other users seemed trustworthy.
5. The operators of the participation platform seemed trustworthy.
9. The website seemed trustworthy.

Trust specific

Community

2. The comments of other users were useful.
3. The other users had good intentions.
4. I could rely on the statements of other users.

Provider

6. The operators of the participation platform have done a good job.
7. It was important to the operators that the users are satisfied with the participation platform.
8. I could rely on the statements of the operators of the participation platform.

Information System

10. The website worked well.
11. I had the impression that my data was necessary for the use of the website.
12. I found the website useful.

Self-efficacy

1. In principle, I felt able to write a comment.
2. I felt technically competent enough to write a comment.
3. In terms of the topic, I felt competent enough to express my opinion.
4. I found the topic too difficult to express my opinion.
5. I found it complicated to write a comment.
6. I was overburdened to write a comment.